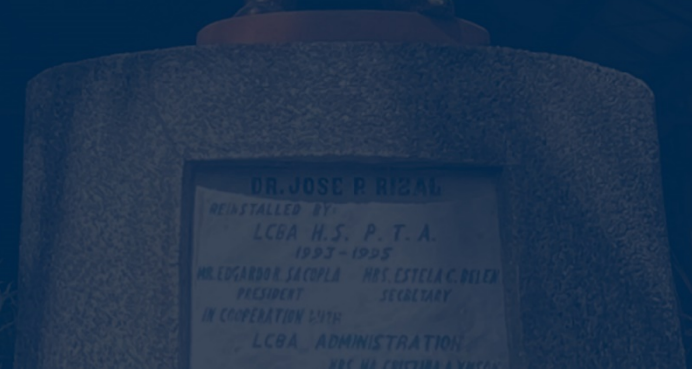




LAGUNA COLLEGE OF BUSINESS AND ARTS
EDUCATION SPANNING GENERATIONS





GENERAL MATHEMATICS





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PRAY





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ATTENDANCE





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Percentages Business and Finance



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Learning Objectives:

1. define percentage-based financial concepts (inflation, mark-up, discount, VAT, profit, and loss) and identify their applications in everyday situations.
 2. solve problems involving percentage increases/decreases, mark-ups, discounts, VAT, profit, and loss using real or simulated data.
-



Learning Objectives:

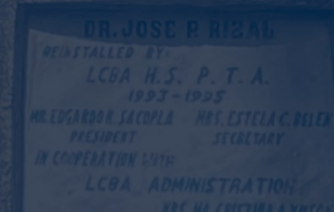
3. analyze and appreciate how inflation, mark-ups, and deductions affect consumer choices, purchasing power, and business profitability, and demonstrate responsible financial decision-making.





Learning Objectives:

4. prepare and present a Business Scenario Analysis (e.g., a simple shop or household budget) that shows the impact of inflation, mark-ups, discounts, VAT, and profit/loss, while demonstrating accuracy, teamwork, and responsibility in completing the task.





An illustration of a white sheet of paper with a checklist. The paper is slightly crumpled and has a yellow pencil resting on it. The checklist consists of five blue horizontal lines, each preceded by a green checkmark. The background is dark blue with faint, stylized palm trees and a building. In the lower-left corner, there is a faint, semi-transparent image of a document or plaque with text including "JOSE P. RIZAL", "1993-1995", "JACUBA", "ESTELA C. DELA", "SECRETARY", and "TOWN SOUTH".





Think (2–3 minutes):

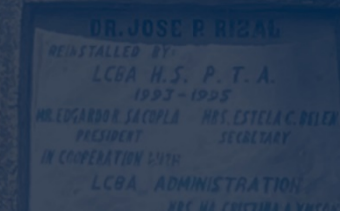
Think of one personal experience related to percentages in real life (examples: discounted clothing, higher jeepney fare, increase in cellphone load prices, allowance increase, seeing VAT on a receipt, or selling an item for profit) and write on your notebook.



Pair (2–3 minutes):

Look for a partner and share your stories with them.

Listen carefully with your partner's sharing and note similarities or differences in your experiences.





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Share (4–6 minutes):

Volunteer to present your experiences to the whole class.

Your teacher will record key points from your sharings.







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Test I. Multiple Choice.

Directions: Choose the letter of the correct answer. Write your answer in CAPITAL LETTERS on the answer sheet.





1. Which of the following best describes “gross pay”?

A. Salary before deductions

B. Salary after deductions

C. Additional benefits only

D. Net income





2. Which government-mandated deduction is intended for retirement benefits of employees?

A. TAX

B. PHILHEALTH

C. PAG-IBIG

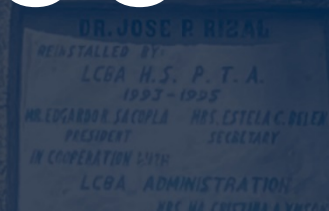
D. GSIS/SSSS





**3. If an employee earns
₱20,000 and deductions total
₱5,000, what is the “net pay”?**

A. ₱25,000 C. ₱15,000
B. ₱20,000 D. ₱5,000





4. Which of the following situations shows the effect of **inflation****?**

A. A cellphone price drops during a sale

B. Grocery prices increase compared to last year

C. A store gives a 10% discount

D. A business earns more profit





5. What does **VAT (Value-Added Tax)**** represent?**

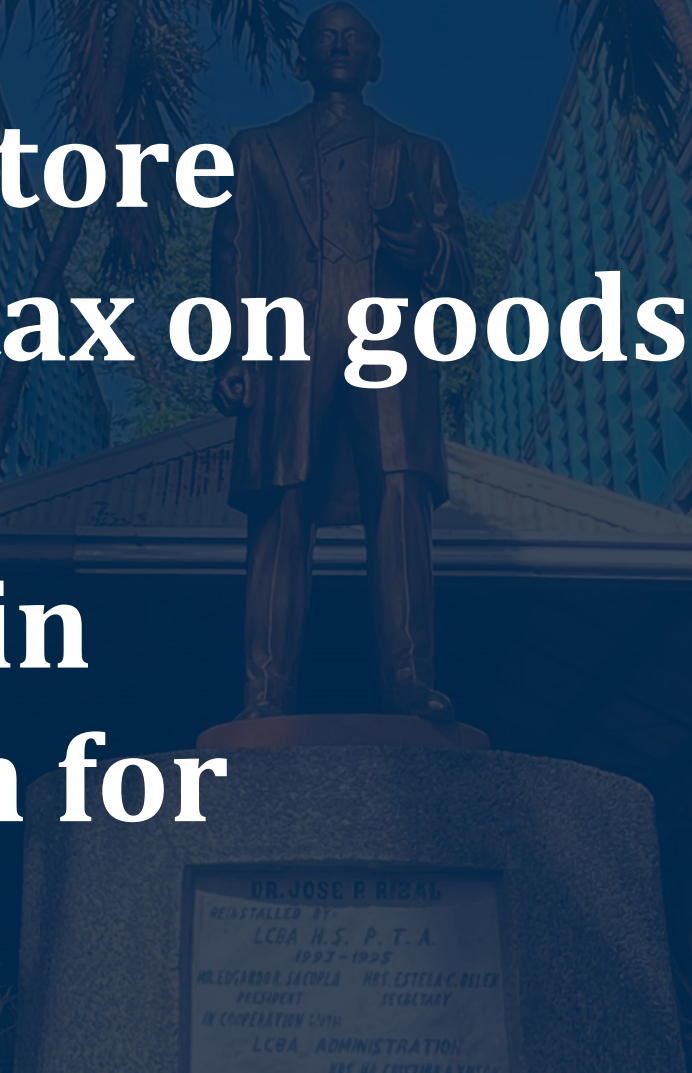
A. A discount given by the store

B. A government-imposed tax on goods and services

C. A company's profit margin

D. An employee's deduction for retirement

.....





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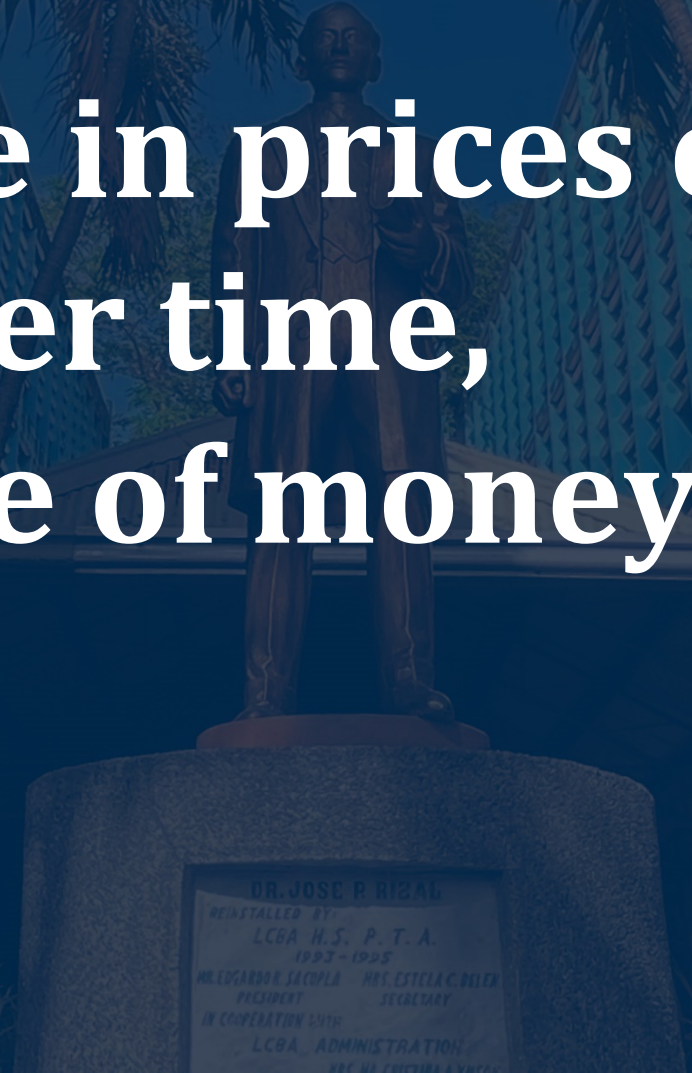
Percentages Business and Finance





Inflation

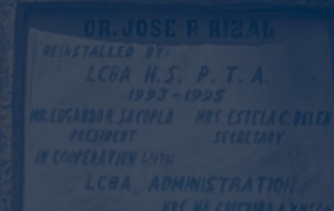
- is the general increase in prices of goods and services over time, which causes the value of money to go down.





Inflation

- is the general increase in prices of goods and services over time, which causes the value of money to go down.





Inflation

- Inflation affects what you can buy with your money, and it is important to know how much things have increased or decreased in price so we can make better financial decisions in life.

The Purchasing Power Gap

Inflation: The general increase in prices causing money's value to drop. A **₱100 burger** last year now costs **₱110**. ↗

Last Year



• ₱1,000

This Year



• ₱1,000

Market Data: Rice
₱40/kilo → **₱48/kilo**

+20%

Market Data: Rice
₱40 → **₱10**

+20%

Market Data: Eggs
₱7 → **₱9**

+28.57%



Illustrative Example 1

The price of 1 kilo of rice was ₱40 last year. This year, it's ₱48. What is the percentage increase?





Solution:

Increase = New Value – Original Value

Increase = $48 - 40 = 8$

Percentage increase = $\left(\frac{\text{increase}}{\text{original value}} \right) \times 100$

Percentage increase = $\left(\frac{8}{40} \right) \times 100 = 20\%$

Thus, the price of rice increased by 20%.





Illustrative Example 2

Your father earned ₱15,000 per month last year. This year, his salary increased to ₱15,750. Meanwhile, inflation was 6% this year. Did your father's salary keep up with inflation?...



Solution:

Increase = New Value – Original Value

Increase = 15,750 – 15,000 = 750

Percentage increase

$$= \left(\frac{\text{increase}}{\text{original value}} \right) \times 100$$

Percentage increase

$$= \left(\frac{750}{15,000} \right) \times 100 = 5\%$$



Markups And Discounts

Term	Definition	Formula
Mark-Up	The amount added to the cost price of an item to make a profit.	Mark-Up = Selling Price – Cost Price
Discount	The amount is reduced from the original price of an item, usually during a sale or promotion.	Discount = Original Price – Net Price
Selling Price	The final price at which the product is sold to the customer after mark-up is added.	Selling Price = Cost Price + Mark-Up
Net Price	The actual amount a buyer pays after the discount has been deducted.	Net Price = Original Price – Discount



Illustrative Example 3

A store buys a bag for ₱500 and adds a 30% mark up to earn profit. What is the selling price?





Solution:

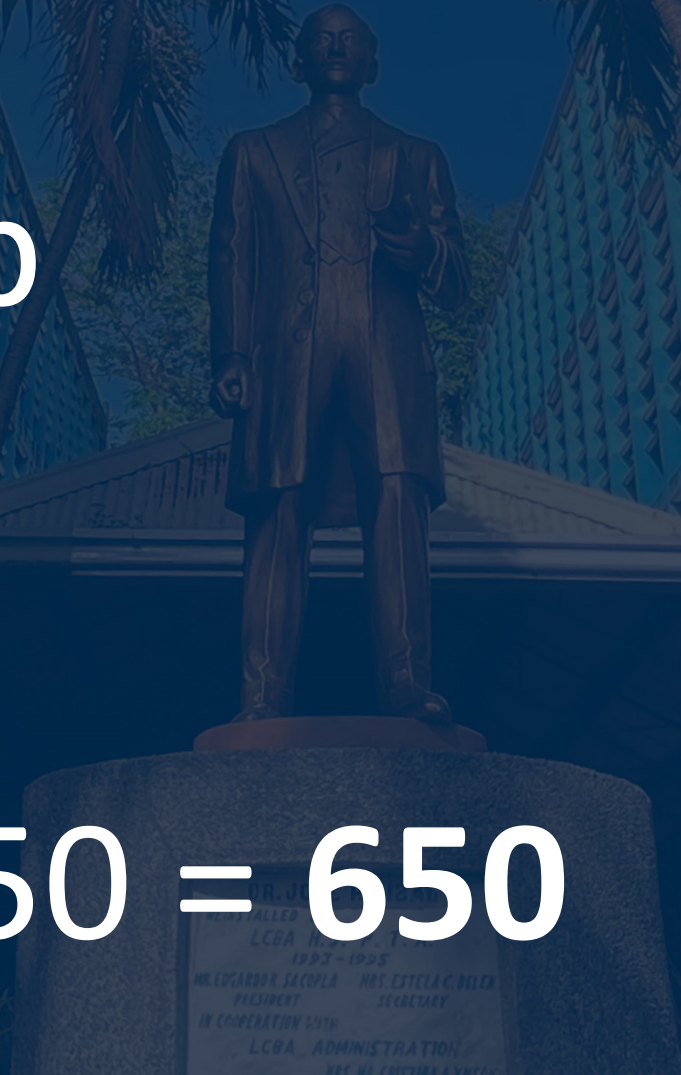
Selling price

= Cost price + Mark-Up

Selling price

= 500 + 30% (500)

Selling price = 500 + 150 = 650
.....





Illustrative Example 4

A ₱1,000 jacket is on 20% discount. What is the net price?





Solution:

Discount
= Original Price – Net Price

Discount
= 1000 – 20% (1000)

Discount = 1000 – 200 = 800
.....



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Illustrative Example 5

An item is bought for ₱400 and sold for ₱520. What is the mark-up percentage?





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Solution:

Mark-Up

= Selling Price – Cost Price

Mark-Up

= 520 – 400 = 120





Solution:

Mark-Up percentage

$$= \left(\frac{\text{Mark-Up}}{\text{Cost Price}} \right) \times 100$$

Mark-Up percentage

$$= \left(\frac{120}{400} \right) \times 100 = 30\%$$

.....





Illustrative Example 6

An original price of ₱2,000 is reduced to ₱1,600. What is the discount percentage?





Solution:

$$\begin{aligned}\text{Discount} &= \text{Original Price} - \text{Net Price} \\ &= 2,000 - 1,600 = 400\end{aligned}$$

$$\begin{aligned}\text{Discount percentage} &= \left(\frac{\text{Discount}}{\text{Original Price}} \right) \times 100 \\ &= \left(\frac{400}{2000} \right) \times 100 = 20\% \\ &\quad \dots\dots\end{aligned}$$





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Value Added Tax (VAT)

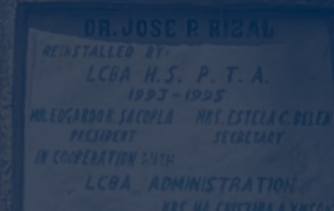


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VAT is applied to everyday purchases such as eating at Jollibee, buying clothes from SM, or shopping online at Lazada or Shopee. VAT in the





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**Philippines is set at 12%, and
it is part of daily life for
consumers, students, parents,
and entrepreneurs.**





Note: A 12% tax is charged on most goods and services in the Philippines. Businesses collect it from customers and pay it to the Bureau of Internal Revenue (BIR)....



Illustrative Example 7

You buy notebooks worth
₱500 (VAT exclusive) at the
National Bookstore. How
much is the VAT and what is
the total price?...



Solution:

$$\text{VAT} = 12\% \times \text{Price}$$

$$\text{VAT} = 12\% (500)$$

$$\text{VAT} = 0.12 (500) = 60$$

$$\text{Total price} = \text{Price} + \text{VAT}$$

$$\text{Total price} = 500 + 60 = 560$$





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Illustrative Example 8

Your total bill at Jollibee is
₱224, already VAT-inclusive.
How much is the VAT?





Solution:

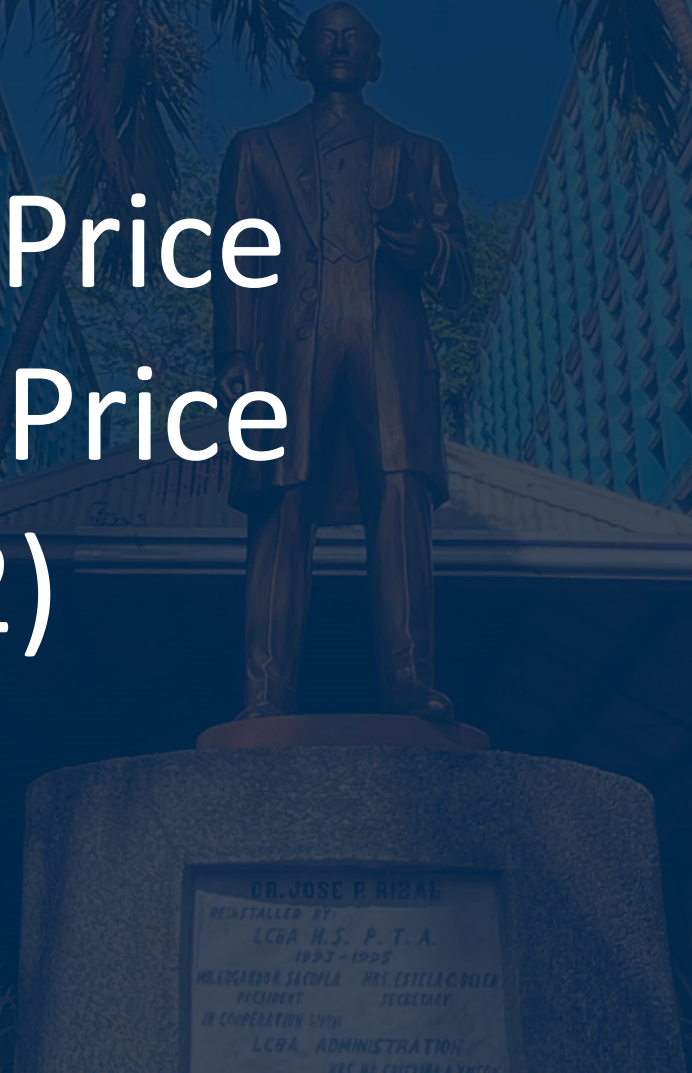
$$\text{Total price} = \text{Price} + \text{VAT}$$

$$\text{Total price} = \text{Price} + 12\% \times \text{Price}$$

$$\text{Total price} = \text{Price} + 0.12 \times \text{Price}$$

$$\text{Total price} = \text{Price} (1 + 0.12)$$

$$\text{Total price} = \text{Price} (1.12)$$





Solution:

Find the Price:

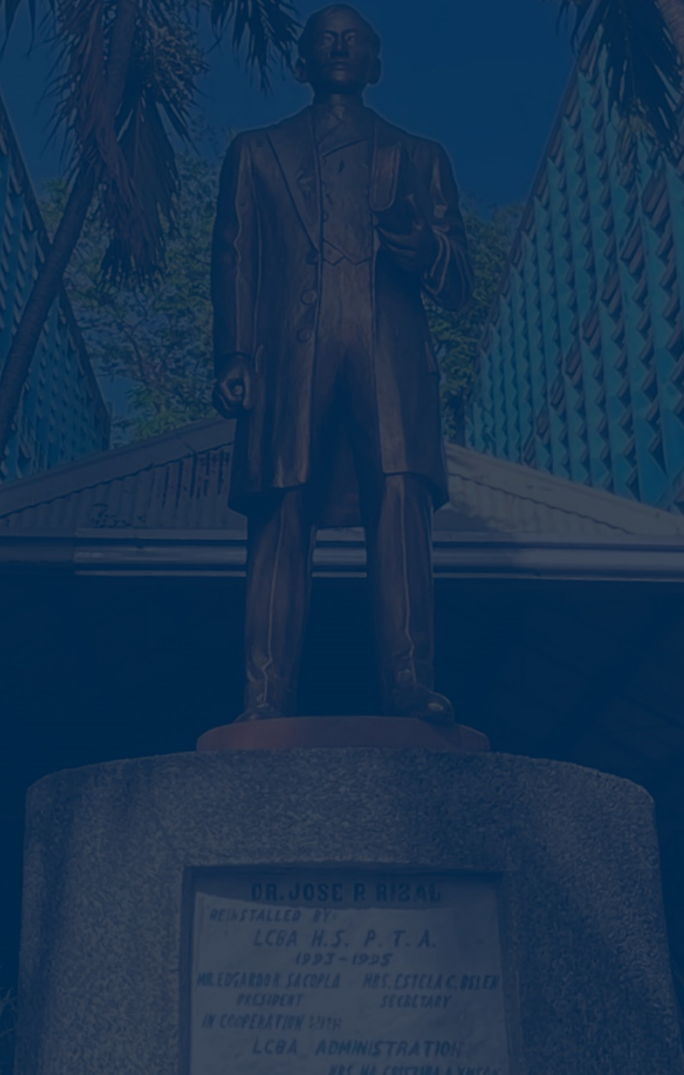
$$224 = \text{Price} (1.12)$$

$$\text{Price} = 224 \div 1.12 = 200$$

Find the VAT:

$$\text{VAT} = \text{Total price} - \text{Price}$$

$$\text{VAT} = 224 - 200 = 24 \dots$$





Illustrative Example 9

You bought a phone case for ₱672 from Shopee, and the price already includes VAT.
What is the price before VAT?





Solution:

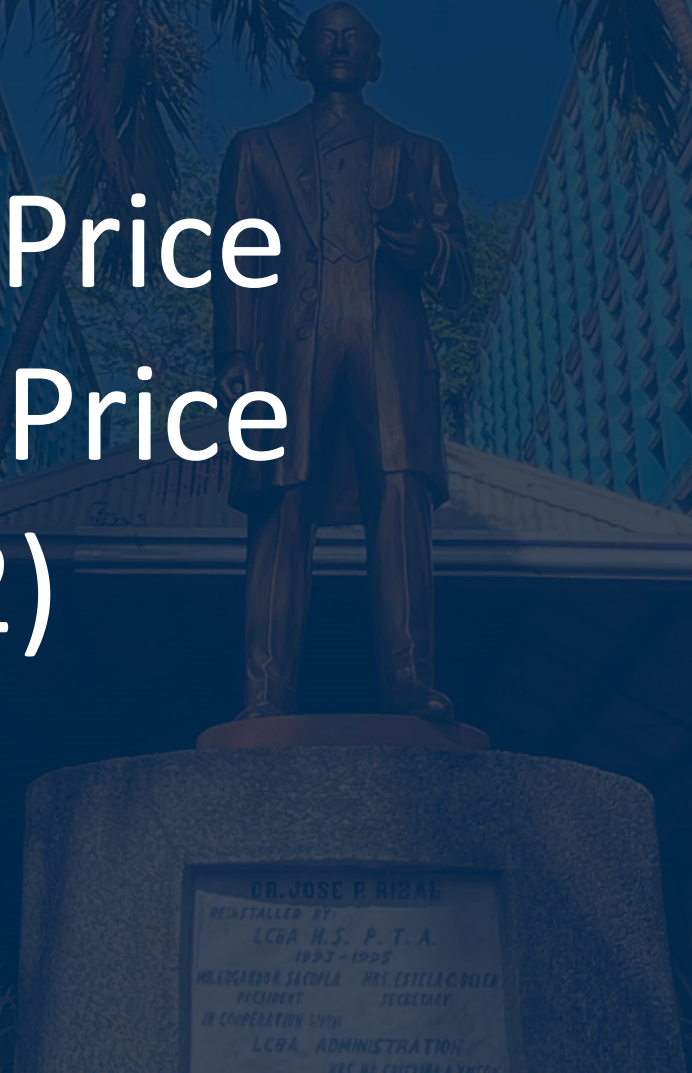
$$\text{Total price} = \text{Price} + \text{VAT}$$

$$\text{Total price} = \text{Price} + 12\% \times \text{Price}$$

$$\text{Total price} = \text{Price} + 0.12 \times \text{Price}$$

$$\text{Total price} = \text{Price} (1 + 0.12)$$

$$\text{Total price} = \text{Price} (1.12)$$





Solution:

Find the Price:

$672 = \text{Price} (1.12)$

$\text{Price} = 672 \div 1.12 = 600$





Illustrative Example 10

You are starting a sari-sari store and want to sell snacks with a ₱1,000 base price. You need to add 12% VAT. What is the final selling price?

.....



Solution:

$$\text{Total price} = \text{Price} + \text{VAT}$$

$$\text{Total price} = \text{Price} + 12\% \times \text{Price}$$

$$\text{Total price} = 1,000 + 0.12 (1,000)$$

$$\text{Total price} = 1,000 + 120 = 1,120$$





Profit and Loss



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In any business transaction, such as running a sari-sari store or an online shop, the goal is to manage the following variables effectively





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In any business transaction, such as running a sari-sari store or an online shop, the goal is to manage the following variables effectively





Cost Price (CP):

This represents your initial investment, including the price paid to buy an item or the expenses incurred to produce it.





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Selling Price (SP): This is the final amount you charge your customers.



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Profit (Gain): This occurs when your revenue exceeds your expenses ($SP > CP$).

Formula:

Profit = Selling Price – Cost Price.





Loss: This occurs when you sell an item for less than what you paid for it ($SP < CP$).

Formula:

Loss = Cost Price – Selling Price





Illustrative Example 11

You bought a hoodie for ₱500 and sold it for ₱620. How much profit did you make? What is the profit percentage?





Solution:

$$\begin{aligned}\text{Profit} &= \text{Selling price} - \text{Cost price} \\ &= 620 - 500 = 120\end{aligned}$$

$$\begin{aligned}\text{Profit percentage} &= \left(\frac{\text{Profit}}{\text{Cost Price}} \right) \times 100 \\ &= \left(\frac{200}{500} \right) \times 100 = 24\%\end{aligned}$$



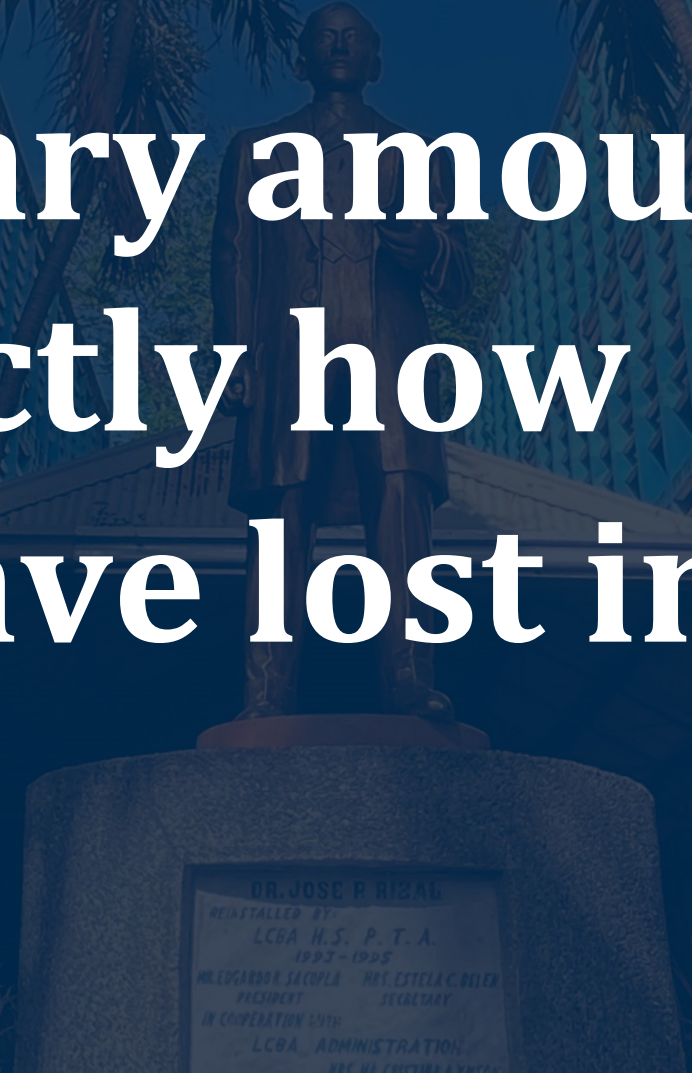




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Absolute Loss

is the actual monetary amount lost. It tells you exactly how much money you have lost in real terms.





Profit Percentage

expresses the loss as a proportion of the original value, shown as a percentage. It provides context for how significant the loss is relative to your initial investment.





- **Formula:**

$$= \left(\frac{\text{Absolute Loss}}{\text{Original Value}} \right) \times 100$$

- **Example:**

$$= \left(\frac{\text{₱ 2,000}}{\text{₱ 10,000}} \right) \times 100$$





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Why both matter?

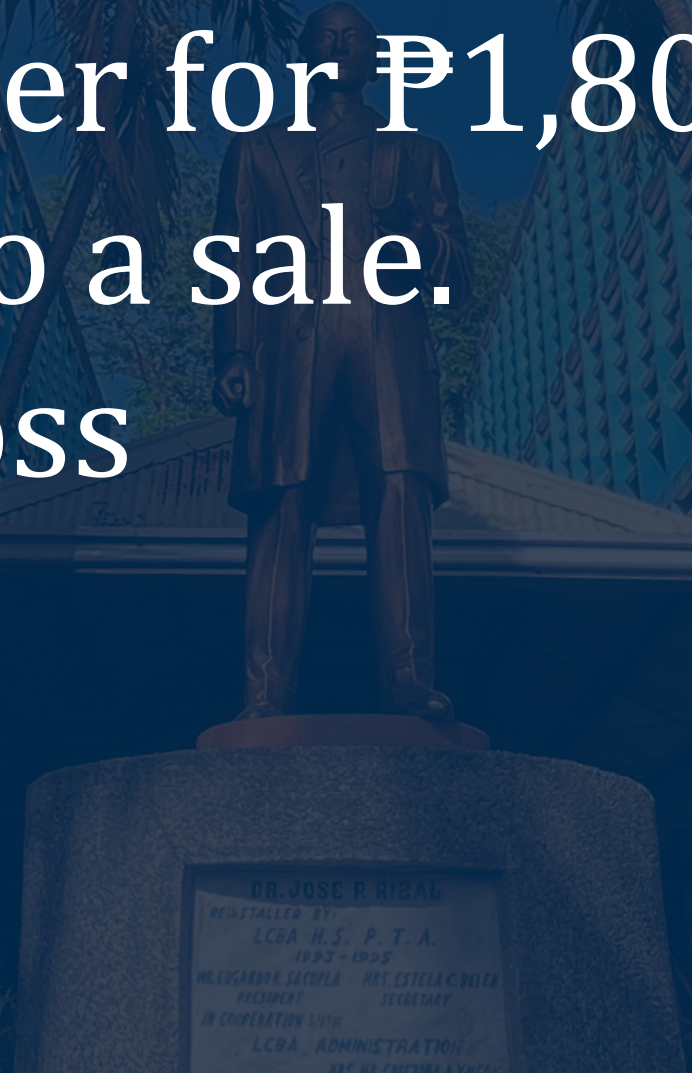


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Illustrative Example 12:

A store bought a rice cooker for ₱1,800 but sold it at ₱1,500 due to a sale. What is the loss and the loss percentage?





Solution:

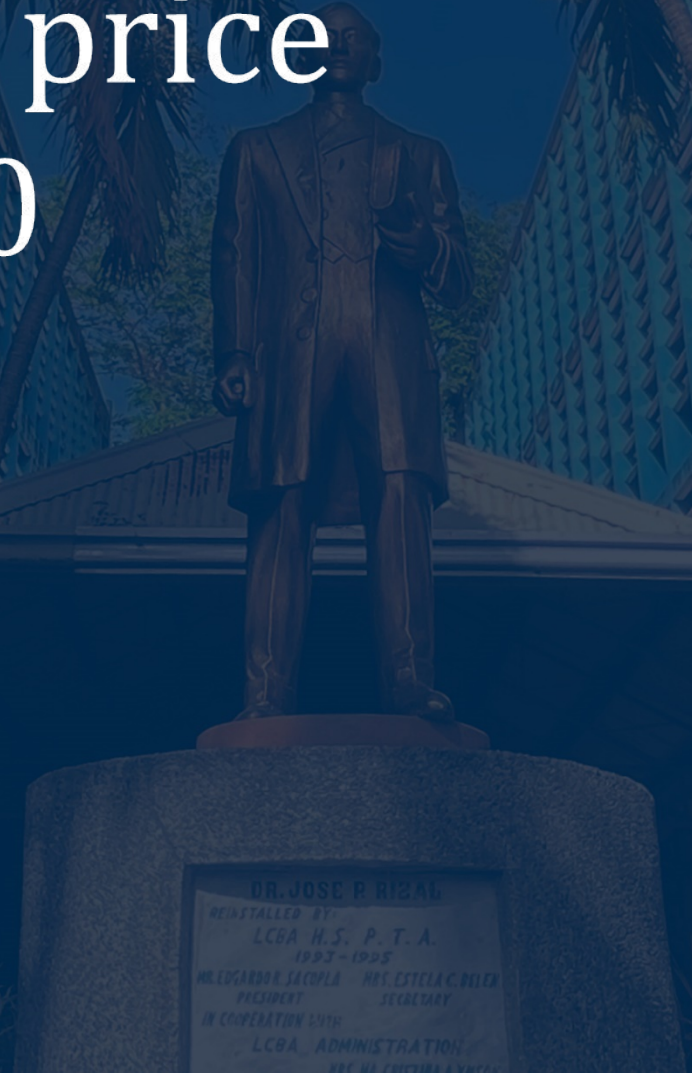
Loss = Cost price – Selling price

Loss = 1,800 – 1,500 = 300

Loss percentage

$$= \left(\frac{\text{Loss}}{\text{Cost Price}} \right) \times 100$$

$$= \left(\frac{300}{1800} \right) \times 100 = 16.67\%$$





Illustrative Example 13:

You want to earn 20% profit on a phone you bought for ₱10,000. What should be the selling price?





Solution:

$$\begin{aligned}\text{Profit} &= \text{Selling price} - \text{Cost price} \\ &= \text{Selling price} - 10,000\end{aligned}$$

$$\text{Profit percentage} = \left(\frac{\text{Profit}}{\text{Cost Price}} \right) \times 100$$

$$20 = \left(\frac{\text{selling Price} - 10,000}{10,000} \right) \times 100$$

$$2,000 = \text{selling Price} - 10,000$$

$$\text{selling Price} = 10,000 + 2,000 = 12,000$$





Illustrative Example 14:

You bought a pair of shoes for ₱1,200. You do not want to lose any money. What is the minimum selling price?



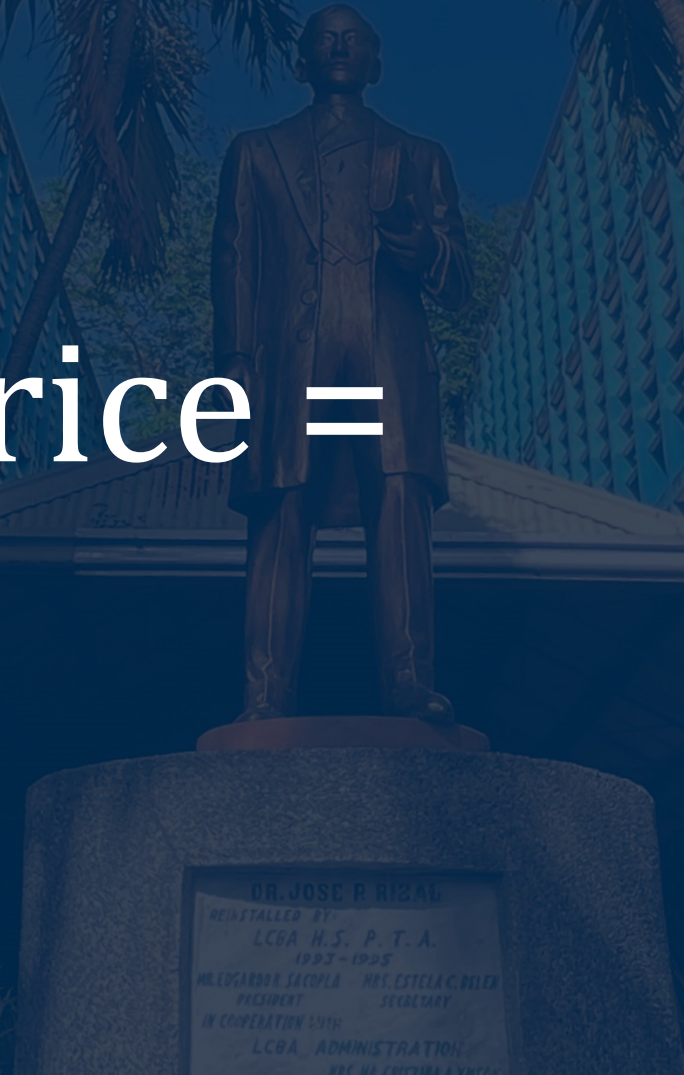


Solution:

To avoid loss,

Selling Price = Cost Price =

₱1,200





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MATH DRILL: PERCENTAGE IN BUSINESS AND FINANCE



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Part 1. Inflation Problems



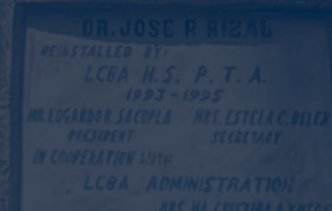


1. The price of eggs
increased from ₱7 to ₱9.
Find the percentage
increase.





2. A family's monthly electricity bill was ₱2,500. Due to a 12% increase in rates, how much is the new bill?





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Part 2. Mark-up Problems



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3. A store buys shoes for ₱1,200 and sells them for ₱1,500. Find the mark-up percentage.





4. A gadget costs ₱8,000. The seller wants a 25% profit. What should be the selling price?





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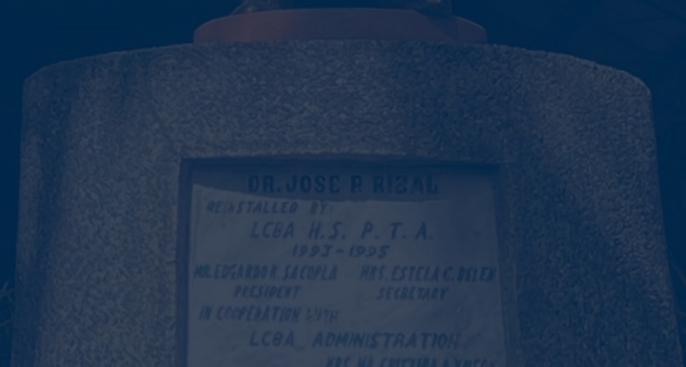
Part 3. Discount Problems



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5. A bag costs ₱3,000.
During a 20% sale, how
much is the discounted
price?





6. A cellphone priced at ₱15,000 is offered at a 10% discount. What is the final price?





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Part 4.

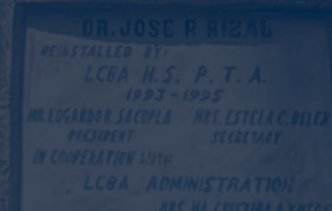
VAT Problems



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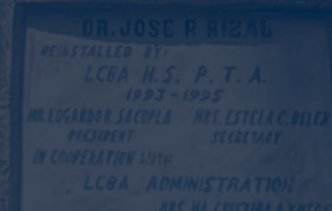


7. A restaurant bill before VAT is ₱1,200. How much is the total with 12% VAT?





8. A laptop costs ₱40,000 including 12% VAT. Find the price before VAT.







9. A trader bought fruits for ₱2,000 and sold them for ₱2,400. Find the profit and profit percentage.





10. A vendor bought umbrellas for ₱5,000 and sold them for ₱4,500. Find the loss and loss percentage.





GENERALIZATION: LESSON WRAP UP!

